Q1

Why does the perceptron algorithm use class labels **−1 and +1** instead of **0 and 1**?
Explain how this choice affects the perceptron update rule.

Q2

Gradient descent usually updates weights by **subtracting** the gradient.
Why does the perceptron update rule **add** the term (eta , y , x) instead?

Q3(optional)

Explain in your own words how the perceptron learning rule can be seen as a form of **gradient descent**.
Also mention one limitation of the perceptron loss function.